Baseball Product — API testing and product validation report

Scope: small-scale functional and manual validation of the Baseball Product against the deployed SportsAPI365 baseball API. This is **not** production certification, load testing, or a security assessment.

Statuses used: `PASS` / `FAIL` / `BLOCKED` / `NOT TESTED` / `NEEDS INVESTIGATION`. A case is marked executed only when it was observed on a live GET. Documented claims that were not independently observed are labeled as documented.

A. Document overview

Field	Value
Product	Baseball Product
App	React UI. No local baseball backend.
API	`GET https://sports-api.sportsdata365.com/msdata/v1/baseball/*`
Catalog	**42** baseball routes
Client	Browser calls a same-origin proxy; keys stay server-side
Config names (values not recorded)	`MSDATA_BASE_URL`, `MSDATA_API_KEY`, `MSDATA_LIVE_KEY`
Nature	Evidence synthesis plus a small set of **additional read-only GETs**. No POST/PUT/DELETE. No load/stress. No app functionality changes.

B. Executive summary

The catalog of **42** baseball routes was executed. **40** returned HTTP 200 on valid entities; **2** catalog routes returned **404** on a valid team and a finished game (`best-players`, team `rankings`). No 5xx responses appeared in the recorded call set (196 calls: 181×200, 8×400, 7×404).

The product can ship **directory, search, match-centre extras, team profile, player profile, and competition total standings**. It cannot ship a trustworthy **Live** or **Results** feed from `GET /events` alone: `status=finished` and `status=inprogress` both returned **`total=0`** while a team `events/last` feed returned **thousands** of games in the same window. Matches → Results therefore falls back to one team's `events/last`.

List rows on `/events` and `/fixtures/today` use a **string** `status` and **`{ series }`** scores. Event **detail** uses a **status object** `{ code, type, description }` and **`inningsBaseball` { run, hits, errors }**. In this review, `/fixtures/today` and `/events` returned the **same event IDs** and the same body size.

Other confirmed gaps: `/h2h` is 200 with `teamDuel`/`managerDuel` null; standings `home`/`away` are 200 with empty `rows` (total has a full table); `page > totalPages` is 200 with an empty list while `total` stays high; `X-Live-Key` alone is 200 and `x-apim-key` alone is 403; football-style **player season statistics** returns 200 for a known player but is **not in the 42-route catalog** and is not called by the player page.

This report does **not** assign numeric experience scores.

C. Product overview + workflow-to-endpoint table

What the product is: a React lab that calls deployed `/v1/baseball/*` through a same-origin `/msdata` proxy. Keys stay in server env.

Routes in the chrome: Home `/`, Matches `/matches`, Competitions `/competitions`, Teams `/teams`, Players `/players`, Search `/search`, "42 routes" `/endpoints`. Additional routes exist but are not in the header: `/match/:id`, `/player/:id`, `/team/:id`, `/manager/:id`, `/competition/:id`.

Feature / screen	Endpoints	Expected	Observed	Finding
Home /	`fixtures/today`, `events`, `tournaments`	Sections Live / Today / Upcoming / Completed + competition cards	Live = events filtered to status label Live; Today = fixtures list; Upcoming/Completed = subsets of today. `/events` and `/fixtures/today` were the same `notstarted` rows. `status=inprogress` total=0 → Live empty. List scores are `series` only.	Live empty; Today cards cannot show R/H/E from the list payload.
Matches → Live	`GET /events?status=inprogress`	Live games	200, `total=0`	Results/Live blocker. `FAIL`
Matches → Today	`GET /fixtures/today`	Today's schedule	200, same IDs as `/events`	Identical to default `/events` in this run. `NEEDS INVESTIGATION` (semantics)
Matches → Upcoming	`GET /events?status=notstarted`	Upcoming	200, non-zero total	Works for the current thin feed. `PASS`

Feature / screen	Endpoints	Expected	Observed	Finding
Matches → Results	`GET /events?status=finished`; if empty, `GET /team/{teamId}/events/last`	Completed games	finished `total=0`; fallback 200 with a large team history. Mixes canceled games into Results.	Fallback is one-team-only. FAIL
Match centre `/match/:id`	`GET /event/{id}` plus tabs: at-bats, lineups, statistics, player-statistics, top-performers, umpires, h2h, best-players, pregame-form, team-streaks, weather, enrichments	Full game	Finished game: detail 200 with R/H/E; best-players **404**; h2h **200 null duels**; other extras 200.	Centre usable except h2h empty and best-players 404.
Search `/search`	`GET /search?q=`	Typed hits → player/team/competition	Known team / player / competition queries returned typed hits. Missing q → 400, `message` is an **array**. Typo / near-miss search: **NOT TESTED**.	Required q works.
Players directory	`GET /players?page&limit=20`	Paginated directory	Large total and many pages. UI pages on `hasMore`.	PASS
Player `/player/:id`	`player/{id}`, `events/last`, `events/next`	Profile + recent/upcoming	Profile 200; `team: null`; last has a large history; next **total=0**. Season stats route not called.	Profile works; no season line; upcoming empty.
Teams directory	`GET /teams`	Directory	Paginated directory with a non-zero total.	PASS
Team `/team/:id`	profile, players, events last/next, featured-event, near-events, performance, rankings, unique-tournaments, standings-seasons, media, enrichments	Team hub	Profile 200; roster present; rankings **404**; events/next total=0; featured-event was a finished game.	Rankings 404 looks like a missing team.
Manager `/manager/:id`	`GET /manager/{id}`	Profile	200 with id, name, imageUrl. Not in header.	API PASS; UI walk for this screen NOT TESTED.
Competitions `/competitions`	`GET /tournaments`	Directory	Paginated directory with a non-zero total.	PASS
Competition `/competition/:id`	unique-tournament, seasons, events, scheduled-events/{date}, standings/{type} total|home|away, round/{round}	Competition hub	Profile 200; seasons and events present; standings total has a full table; home/away **0** rows.	Home/away tables empty.
42 routes `/endpoints`	Every catalog path	Click-to-GET lab	Lab page exists; catalog sweep covers the same paths.	Lab page exists; sweep PASS as API coverage.

D. API inventory (42 catalog routes + extra observed)

Base path: `https://sports-api.sportsdata365.com/msdata` + `/v1/baseball/...`. Auth headers used: `x-apim-key` (from `MSDATA_API_KEY`) and `X-Live-Key` (from `MSDATA_LIVE_KEY`).

D1. Catalog (42)

#	Route	HTTP	Kind / list key	Count / pagination	Inventory note
1	`/event/{eventId}`	200	match `event`	—	Detail. Status object; `inningsBaseball`.
2	`/event/{eventId}/at-bats`	200	`atBats`	paginated	Empty 200 on canceled/not-started.
3	`/event/{eventId}/best-players`	**404**	error envelope	—	`ENRICHMENT_NOT_FOUND` on a finished game.
4	`/event/{eventId}/enrichments`	200	`types`	several types	Includes h2h_events, highlights, series, team_streaks, votes.
5	`/event/{eventId}/enrichments/{type}`	200	`events`	paginated	
6	`/event/{eventId}/h2h`	200	`teamDuel`, `managerDuel`	both **null**	
7	`/event/{eventId}/lineups`	200	`home`, `away`	dressed players present; `manager` null on one side	
8	`/event/{eventId}/player-statistics`	200	`playerStatistics`	paginated	
9	`/event/{eventId}/player/{playerId}/statistics`	200	player object	—	Match-level, not season.
10	`/event/{eventId}/pregame-form`	200	`homeTeam`, `awayTeam`	form arrays + eventIds	
11	`/event/{eventId}/statistics`	200	`statistics`	period ALL	Fielding + batting groups.
12	`/event/{eventId}/team-streaks`	200	six keys, all `[]`	—	Not-started game 404 in lab.
13	`/event/{eventId}/top-performers`	200	`topPerformers`	several, roles batter	
14	`/event/{eventId}/umpires`	200	`umpires`	crew present	HP/1B/2B/3B/LF/RF.

#	Route	HTTP	Kind / list key	Count / pagination	Inventory note
15	`/event/{eventId}/weather`	200	`conditions`	present	
16	`/events`	200	`items`	small `total`; paginated	List shape ≠ detail.
17	`/fixtures/today`	200	`items`	same total as `/events`	Same IDs as `/events` in this review.
18	`/manager/{managerId}`	200	`manager`	id, name, imageUrl	
19	`/player/{playerId}`	200	`player`	`team: null`	
20	`/player/{playerId}/events/{direction}`	200	`events`	last: large total; next: total=0	
21	`/players`	200	`items`	large total, many pages	
22	`/search`	200	`items`	typed hits	type/id/name.
23	`/team/{teamId}`	200	`team`	`country: null`; logoUrl present	
24	`/team/{teamId}/enrichments`	200	`types`	several types	
25	`/team/{teamId}/enrichments/{type}`	200	`events`	paginated	
26	`/team/{teamId}/events/{direction}`	200	`events`	last: thousands of games; next: total=0	
27	`/team/{teamId}/featured-event`	200	match `event`	finished game	Not “currently live”.
28	`/team/{teamId}/media`	200	`media`	paginated	
29	`/team/{teamId}/near-events`	200	`previousEvent`, `nextEvent`	—	
30	`/team/{teamId}/performance`	200	`events` (+ form, points)	paginated	
31	`/team/{teamId}/players`	200	`players`	roster present	Nested `player` objects.
32	`/team/{teamId}/rankings`	**404**	error envelope	—	`ENRICHMENT_NOT_FOUND`.
33	`/team/{teamId}/standings-seasons`	200	`uniqueTournamentSeasons`	paginated	
34	`/team/{teamId}/unique-tournaments`	200	`uniqueTournaments`	present	
35	`/teams`	200	`items`	paginated	
36	`/tournaments`	200	`items`	paginated	
37	`/unique-tournament/{tournamentId}`	200	`uniqueTournament`	competition object	`currentSeasonId` missing on object; seasons envelope has it.
38	`/unique-tournament/{tournamentId}/events`	200	`events`	very large total	Deep page numbers return real data.
39	`/unique-tournament/{tournamentId}/scheduled-events/{date}`	200	`events`	day’s events	
40	`/unique-tournament/{tournamentId}/season/{seasonId}/round/{round}`	200	`events`	large total	
41	`/unique-tournament/{tournamentId}/season/{seasonId}/standings/{type}`	200	`standings`	total: full table; home/away: 0 rows	
42	`/unique-tournament/{tournamentId}/seasons`	200	`seasons`	paginated	`currentSeasonId` on envelope.

D2. Extra observed routes / calls labeled **not-in-catalog**

These are **not** in the 42-route baseball catalog. They were still observed.

Path	HTTP	Label
`/v1/baseball/player/{playerId}/unique-tournament/{tournamentId}/season/{seasonId}/statistics`	**200**	**not-in-catalog**. Keys include `player`, `team`, `statistics`, `playerId`, `teamId`, `uniqueTournamentId`, `seasonId`. Nested batting stats (average, home runs, at-bats, appearances, hits).
`/player/{id}/events/sideways`	400	Invalid `direction` on a catalog template, not a new route.
`/unique-tournament/.../standings/form`	400	Invalid standings `type`.
`/unique-tournament/.../scheduled-events/not-a-date`	400	Invalid date.
`/event/abc`, `/event/{unknown}`, `/player/{unknown}`, `/team/{unknown}`	400 / 404	Invalid/unknown IDs on catalog templates.

Basketball-only `incidents`, `injuries`, and season-statistics paths are **absent from the baseball catalog**. Season statistics is nonetheless **reachable** (row above). Incidents/injuries were **not** called in this review (`NOT TESTED`).

E. Response schema and data quality

Truncated previews are not treated as full payloads; event-detail GETs are used for score/status fields.

E1. Event list vs event detail (confirmed)

`GET /events` and `GET /fixtures/today` list row shape:

```
{
  "id": "<eventId>",
  "status": "notstarted",
  "homeScore": { "series": "<n>" },
  "awayScore": { "series": "<n>" },
  "tournamentId": "<tournamentId>",
  "competitionName": "<name>",
  "seasonId": "<seasonId>"
}
```

Row keys: `id, tournamentId, homeTeam, awayTeam, startTimestamp, status, homeScore, awayScore, seasonId, competitionName`. No `tournament` object. `status` is a **string**. `startTimestamp` is unix seconds.

`GET /event/{eventId}` for a finished game:

- Teams, venue present.
- `status`: `{ "code": 100, "type": "finished", "description": "Ended" }`; `statusType`: `"finished"`.
- `homeScore` keys include `series, current, display, innings, period1–period9, normaltime, inningsBaseball`.
- `homeScore.inningsBaseball`: `{ "run": "<n>", "hits": "<n>", "errors": "<n>" }` (same keys on away).
- `tournament.id`: **null**; `tournament.name` present; nested `uniqueTournament.id` present.
- `startTimestamp`: integer unix seconds.

The client mapping reads `uniqueTournament` **or** `tournament` **or** nested uniqueTournament. Because `tournament` exists, `tournament.id` (null) wins — Match → Competition link can fail even though a nested competition id is present.

Team `events/last` uses list key **`events`** (not `items`) and a **status object**. First row in this review was **canceled**, with `homeScore` keys `["series"]` only.

E2. List-key split (observed)

Family	Array key	Pagination fields
`/events`, `/fixtures/today`, `/teams`, `/players`, `/tournaments`, `/search`	`items`	`page, limit, total, totalPages, hasMore`
team/player/unique-tournament event lists, at-bats, media, etc.	`events` / `atBats` / `media` / …	same pagination fields when present
team roster	`players`	no page fields in probe
standings	`standings[].rows`	no page fields

E3. Other shapes (sanitized)

Resource	Observed
Search hit	`{ "type": "team", "id": "<id>", "name": "<name>", "countryAlpha2": null, "gender": "M" }`
H2H	`{ "teamDuel": null, "managerDuel": null }`
Team streaks (finished game)	all six buckets `[]`
Top performers	several `role: "batter"` with battingHits / battingAtBats / battingHomeRuns / battingRbi
Umpires	assignment HP, 1B, 2B, 3B, LF, RF
Weather	temperature, humidity, windSpeed, isForecast true/false
Pregame form	`form` W/L letters; eventIds listed; `source: "derived"`
Lineups	`confirmed`, `startersListed`, `dressedPlayers`, `manager` (null on one side), `formation: null`; player rows have `battingOrder`, `appearanceRoles`, `startingPitcher`
Player	position, jersey, height present; `team: null`
Team	`country: null`, `manager: null`, `logoUrl` present (path only)
Standings total row keys	team, position, matches, wins, draws, losses, scoresFor/Against, gamesBehind, winPercentage, group, …
Error envelope	`statusCode, message, error, path, timestamp, requestId` (+ `details` on 404)
Validation `message`	sometimes **string**, sometimes **array of strings** (search missing q; `status=ended`)

E4. Quality issues

- A finished game has a full baseball box on **detail**, not on **list**.
- Player `team` is null on the player object.

- Team `country` is null on `/team/{id}` while the same club nested on event detail has `country: { name, alpha2 }`.
- Shirt numbers on lineup preview rows shown are null.
- Client call logs use unix-ms timestamps; API `startTimestamp` is unix **seconds**.

F. Manual API test strategy

Goal: confirm the 42 catalog routes respond, that product screens can consume them, and that known gaps still reproduce. Not a performance SLA, not a soak test, not auth-bypass research beyond the four-header matrix already documented.

In scope

- GET against `/v1/baseball/*`.
- Happy path on IDs found via search → team events → lineups → seasons.
- Status filters, pagination edges, validation 400/404.
- Product workflows on the main screens.

Out of scope

- Load/stress, fuzzing, destructive writes (proxy is documented as **405** for non-GET/HEAD — **not executed** here).
- Sports other than baseball (proxy rejects paths that are not `/v1/baseball`).
- Certifying production SLOs or completeness of global baseball coverage.

How tests were executed

1. Discover IDs via search, team `events/last`, lineups, seasons, enrichments; call each of 42 templates.
2. Extra probes: finished/inprogress, far page numbers, search missing q, invalid/unknown ids, standings home/away, player/team `events/next`.
3. Variation tests (states, second team/player, deep pages, limits, status tokens, enrichment types).
4. Full-catalog lab sweep and in-product HTTP log.
5. Additional **read-only GETs** (auth matrix, identity of fixtures vs events, filters, detail R/H/E, h2h/best-players/top-performers/rankings, standings, far player pages, search without q, not-in-catalog season stats).

Environments: probes hit `MSDATA_BASE_URL` directly with both key headers. The UI hits the same-origin `/msdata` proxy (which injects the same header names). Additional GETs used both headers except the auth matrix.

G. Detailed manual test cases

Legend: **Executed** = observed on a live GET. **Expected** is the product/catalog expectation, not an SLA.

G1. Catalog routes (42)

ID	Endpoint / workflow	Scenario	Steps	Expected	Actual	Status	Notes
TC-C-01	`GET /event/{eventId}`	Finished game detail	GET	200 match with teams, status, scores	200; status object finished; inningsBaseball present	PASS	Nested uniqueTournament id present; `tournament.id` null
TC-C-02	`GET /event/{id}/at-bats`	Play-by-play	GET page=1	paginated at-bats	200, paginated	PASS	Canceled/not-started: empty 200
TC-C-03	`GET /event/{id}/best-players`	Optional MOTM	GET	usable list or empty 200	**404** ENRICHMENT_NOT_FOUND	FAIL	Catalog includes the route
TC-C-04	`GET /event/{id}/enrichments`	Type index	GET	types[]	200, several types	PASS	
TC-C-05	`GET /event/{id}/enrichments/{type}`	h2h_events	GET	paged events	200, paginated	PASS	Other listed types also 200
TC-C-06	`GET /event/{id}/h2h`	Prior meetings	GET	duels or documented empty	200, both duels **null**	FAIL	HTTP 200 is not an error; product H2H tab is blank
TC-C-07	`GET /event/{id}/lineups`	Roster	GET	home/away players	200; dressed players present; manager null on one side	PASS	Baseball roster, not XI=11
TC-C-08	`GET /event/{id}/player-statistics`	Box	GET	paged stats	200, paginated	PASS	

ID	Endpoint / workflow	Scenario	Steps	Expected	Actual	Status	Notes
TC-C-09	`GET /event/{id}/player/{playerId}/statistics`	One player match stats	GET	200 stats	200	PASS	Not season stats
TC-C-10	`GET /event/{id}/pregame-form`	Form	GET	W/L form	200	PASS	
TC-C-11	`GET /event/{id}/statistics`	Team comparison	GET	groups	200, period ALL, Fielding+Batting	PASS	
TC-C-12	`GET /event/{id}/team-streaks`	Streaks	GET	streak buckets	200, all arrays empty	NEEDS INVESTIGATION	Not-started 404 in lab
TC-C-13	`GET /event/{id}/top-performers`	Batters/pitchers	GET	list	200, several batters	PASS	Usable MOTM stand-in
TC-C-14	`GET /event/{id}/umpires`	Crew	GET	umpires	200	PASS	
TC-C-15	`GET /event/{id}/weather`	Conditions	GET	conditions	200	PASS	
TC-C-16	`GET /events`	Default feed	GET page=1	current matches	200, small total, all notstarted in this run	PASS	Business completeness vs team history is TC-X-01
TC-C-17	`GET /fixtures/today`	Today	GET	today's games	200; **same IDs as /events**	NEEDS INVESTIGATION	Identical feed in this run
TC-C-18	`GET /manager/{managerId}`	Profile	GET	profile	200 name + imageUrl	PASS	
TC-C-19	`GET /player/{playerId}`	Profile	GET	profile	200, team null	PASS	Thin preferred fields
TC-C-20	`GET /player/{id}/events/last`	Recent	GET	paged events	200, large total	PASS	
TC-C-21	`GET /players`	Directory	GET	paged players	200, large total	PASS	
TC-C-22	`GET /search`	Known team name	GET	typed hits	200, team hit	PASS	
TC-C-23	`GET /team/{teamId}`	Profile	GET	profile	200, country null	PASS	
TC-C-24	`GET /team/{id}/enrichments`	Index	GET	types	200	PASS	
TC-C-25	`GET /team/{id}/enrichments/{type}`	events_last	GET	payload	200, paginated	PASS	
TC-C-26	`GET /team/{id}/events/last`	History	GET	paged events	200, **thousands** of games	PASS	Contrasts with /events?status=finished
TC-C-27	`GET /team/{id}/featured-event`	Featured	GET	one event	200; finished game	PASS	Semantics unclear vs live
TC-C-28	`GET /team/{id}/media`	Media	GET	paged media	200	PASS	
TC-C-29	`GET /team/{id}/near-events`	Prev/next	GET	envelope	200	PASS	
TC-C-30	`GET /team/{id}/performance`	Form	GET	form+events	200	PASS	Scope undocumented
TC-C-31	`GET /team/{id}/players`	Roster	GET	players[]	200, roster present	PASS	Nested player id/name

ID	Endpoint / workflow	Scenario	Steps	Expected	Actual	Status	Notes
TC-C-32	`GET /team/{id}/rankings`	Rankings	GET	list or empty 200	**404** ENRICHMENT_NOT_FOUND	FAIL	
TC-C-33	`GET /team/{id}/standings-seasons`	Seasons with tables	GET	list	200	PASS	
TC-C-34	`GET /team/{id}/unique-tournaments`	Team competitions	GET	list	200	PASS	
TC-C-35	`GET /teams`	Directory	GET	paged teams	200	PASS	
TC-C-36	`GET /tournaments`	Directory	GET	paged competitions	200	PASS	
TC-C-37	`GET /unique-tournament/{id}`	Competition	GET	competition	200	PASS	currentSeasonId missing on object
TC-C-38	`GET /unique-tournament/{id}/events`	Competition feed	GET	paged events	200, very large total	PASS	
TC-C-39	`GET /unique-tournament/{id}/scheduled-events/{date}`	Calendar day	GET	day's events	200	PASS	
TC-C-40	`GET /unique-tournament/{id}/season/{sid}/round/{round}`	Round 1	GET	events	200	PASS	
TC-C-41	`GET /unique-tournament/{id}/season/{sid}/standings/total`	Table	GET	rows	200, full table, type total	PASS	
TC-C-42	`GET /unique-tournament/{id}/seasons`	Season list	GET	seasons	200	PASS	

G2. Filters, pagination, errors, auth, extra route

ID	Endpoint / workflow	Scenario	Steps	Expected	Actual	Status	Notes
TC-X-01	`GET /events?status=finished`	Results filter	GET	non-zero when history exists	200, items=[], **total=0**	FAIL	Team last feed has thousands in the same window
TC-X-02	`GET /events?status=inprogress`	Live filter	GET	live games or honest empty if none live	200, **total=0**	FAIL	Cannot distinguish "no live games" vs broken index
TC-X-03	`GET /events?status=notstarted`	Upcoming filter	GET	upcoming	200, non-zero total	PASS	
TC-X-04	`GET /events?status=canceled`	Canceled filter	GET	canceled games	200, total=0	FAIL	Canceled games exist on team last
TC-X-05	`GET /events?status=postponed`	Postponed	GET	postponed or empty	200, total=0	NEEDS INVESTIGATION	No independent postponed id confirmed
TC-X-06	`GET /events?status=ended`	Invalid token	GET	400	400; allowed: notstarted, inprogress, halftime, pause, finished, postponed, canceled, awarded, interrupted, coverage_lost	PASS	ended is not in the enum; detail `description` is "Ended"
TC-X-07	`GET /events` page 1 vs 2	Paging	GET page=1 then 2	different slices	different first-row ids	PASS	
TC-X-08	`GET /events?page=9999`	Beyond totalPages	GET	honest empty	200, items=[], **total unchanged**, hasMore=false	FAIL	Empty list is honest; leftover total is not
TC-X-09	`GET /players?page=9999`	Beyond last page	GET	honest empty	200, n=0, **total unchanged**	FAIL	
TC-X-10	`GET /players?page=2000`	Deep page vs last page	GET	empty 200	200, count=0	PASS	No fake rows
TC-X-11	`GET /unique-tournament/{id}/events?page=2000`	page 2000 in range	GET	real rows	200, real rows, large total	PASS	Page 2000 is valid here

ID	Endpoint / workflow	Scenario	Steps	Expected	Actual	Status	Notes
TC-X-12	`GET /events?limit=2000`	Oversize limit	GET	validation error	**400**	PASS	Same for search limit=2000
TC-X-13	`GET /events?limit=1` / `limit=100`	Limit honoured	GET	n≤limit	limit=1 → n=1; large limit capped at total	PASS	
TC-X-14	`GET /search` missing q	Validation	omit q	400	400, message **array** of strings	PASS	
TC-X-15	`GET /event/abc`	Non-numeric id	GET	400	400 string message	PASS	
TC-X-16	`GET /event/{unknown}`	Unknown id	GET	404	404 EVENT_NOT_FOUND	PASS	Contrast with enrichment 404
TC-X-17	Standings home/away	Split tables	GET home, away	rows or documented empty	200, 1 table each, **rows: []**; total has a full table	FAIL	
TC-X-18	Player/team `events/next`	Upcoming	GET	upcoming or empty	both 200, total=0	PASS	Empty may be off-season; not a transport error
TC-X-19	Auth: both headers	`/events?limit=1`	GET	200	200, keys items/page/limit/total	PASS	Values not recorded
TC-X-20	Auth: `X-Live-Key` only	same	GET	documented contract unknown	**200**	PASS	
TC-X-21	Auth: `x-apim-key` only	same	GET	unknown	**403** Forbidden JSON	FAIL	Fail vs “either key works”; contract missing
TC-X-22	Auth: no key	same	GET	401/403	**403**, body not JSON	PASS	Non-JSON error vs JSON 403 with APIM only
TC-X-23	Season stats **not-in-catalog**	Known player + competition + season	GET football-style path	unknown / catalog says absent	**200**, batting average / HR / AB / appearances	PASS	Not wired on the player page
TC-X-24	Invalid direction `events/sideways`	Enum	GET	400	400 “enum string is expected”	PASS	
TC-X-25	Invalid standings `form`	Enum	GET	400	400	PASS	
TC-X-26	Invalid calendar date	`not-a-date`	GET	400	400 expected YYYY-MM-DD	PASS	
TC-X-27	Event states: canceled / notstarted	Same sub-routes	GET detail, lineups, statistics, at-bats, top-performers, best-players	documented empty vs 404	detail/lineups/statistics/top-performers 200; at-bats empty 200; best-players 404 all three states	PASS	Empty vs unsupported still undocumented
TC-X-28	Second entities	Another team / player last	GET events/last	200	200, large totals	PASS	

G3. Product workflows (UI)

ID	Endpoint / workflow	Scenario	Steps	Expected	Actual	Status	Notes
TC-P-01	Home Live	Open /	Load Home	Live cards if games in progress	Code filters events to Live; API inprogress total=0 → empty “No matches.”	FAIL	
TC-P-02	Home Today cards	Open /	View Today	R/H/E + competition name	List series scores; competitionName string unused by card mapping	FAIL	
TC-P-03	Matches Results fallback	Open Results	Click Results	Completed games	`/events?status=finished` empty then one-team `events/last` fallback	FAIL	One team only
TC-P-04	Match centre	Open a finished match	Open tabs	Header + extras	Detail 200; best-players tab HTTP 404; h2h null JSON	FAIL	FAIL is for best-players/h2h tabs; overview PASS (see C-01)

ID	Endpoint / workflow	Scenario	Steps	Expected	Actual	Status	Notes
TC-P-05	Search known team	`/search`	Submit a known team name	Team link	200, type team	PASS	
TC-P-06	Search player / competition	`/search`	Submit	Player / competition	Catalog audit 200. Search page grouping of those queries `NOT TESTED`	PASS	
TC-P-07	Search typo	Near-miss query	Submit	empty or near-miss	Not observed	NOT TESTED	
TC-P-08	Player page	Open a known player	Open	Profile + recent	Overview 200. Season stats not requested by page	PASS	Season line absent by catalog, not by API (TC-X-23)
TC-P-09	Team rankings tab	Open rankings	Open tab	Rankings or empty	GET 404	FAIL	
TC-P-10	Competition standings home/away	Toggle home/away	Toggle	Tables	API empty rows	FAIL	total PASS (TC-C-41)
TC-P-11	Manager from lineup	Open manager	Open	Profile	API 200. No recorded UI walk	NOT TESTED	Header has no Manager link
TC-P-12	Endpoint lab	`/endpoints`	Click each path	42 GETs	Page exists; covered at API layer, not as this page	NOT TESTED	
TC-P-13	Proxy methods	`/msdata`	Non-GET	405	Documented on the proxy; not executed	NOT TESTED	Non-destructive policy

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H. Pagination analysis

Observed pagination fields: `page`, `limit`, `total`, `totalPages`, `hasMore`. Not every list uses them (roster `players[]`, standings, umpires, top-performers).

Observation	Result
Page 2 is a new slice on `/events`	PASS
Directories page 2 differ	PASS
`page > totalPages` → HTTP **200**, empty array, **`total` unchanged**, `hasMore: false`	FAIL as a client-stop contract; PASS that rows are not fabricated
A deep page **inside** range returns real rows	PASS
`limit=2000` rejected	PASS
`limit=1` and a large valid `limit` honoured	PASS
Search huge page: total stays, items empty	same leftover-total pattern
UI paging uses `hasMore` only	Safe against leftover `total`, but only if `hasMore` is false (it was on far pages)

I. Error handling and HTTP status

HTTP	When observed	Body shape	Product handling
200	Most catalog GETs	Resource JSON	Parsed as JSON; treated as error only if `!response.ok`
200 empty	filters total=0; h2h nulls; events/next; standings home/away rows=[]	Valid JSON	Matches Results treats empty finished as fallback; Home Live shows “No matches.” H2H tab still 200 so the panel shows nulls, not an error
400	search missing/short `q`; non-numeric event id; events status=ended; events/sideways; standings/form; invalid date; limit=2000	`error: "Bad Request"`; `message` string **or** array	Search shows concatenated/stringified message
403	`x-apim-key` only; no key	APIM-only: JSON Forbidden; no-key: **non-JSON**	UI copy for 401/403 mentions setting both key names — does not say Live-Key-only is sufficient
404 entity	unknown event/player/team numeric id	`details.code` EVENT_NOT_FOUND / equivalent	Distinguishable message
404 enrichment	best-players, rankings; team-streaks on not-started	`details.code` ENRICHMENT_NOT_FOUND, `reason` no_data_recorded	Same HTTP as missing entity; Match/Team tabs look broken
404	(none for a missing proxy)	—	Client maps 404 to “proxy not found” **even for API enrichment 404s** — product copy can mis-blame the proxy
405	non-GET/HEAD on proxy	text “Method not allowed”	Documented only; not executed
5xx	—	—	**Not observed**

Match/Team tabs always render raw JSON for extras rather than mapping 404 vs empty.

J. Product-level manual validation

Method: source review of every product route, plus recorded UI screens (Home, Full catalog, Match Centre, Team, Competition, Matches, Search, Player). Manager and Endpoint lab page were not walked as named screens.

Journey	Can a user complete it?
Browse today's slate	Partially: upcoming cards, series scores, competition name on the list row
See live games	No — Live empty while filter returns total=0
See completed games globally	No — Results empty then one-team fallback
Open a finished game and read R/H/E	Yes on a finished game
At-bats / lineups / box / umpires / weather / top-performers	Yes on a finished game
Best players tab	No — 404
H2H tab	No useful data — null duels
Search a team and open it	Yes
Open a player and see season AVG/HR	No in UI (route exists off-catalog)
Team roster	Yes
Team rankings	No — 404
Competition total table	Yes
Competition home/away tables	No — empty rows
Manager profile	API yes; chrome does not link it
Click all 42 routes	Page exists; API already swept

Consistency across pages: list feeds without a tournament object show "Competition unknown" when `tournament.id` and `name` are missing; fixtures rows have `competitionName` but the client mapping reads `uniqueTournament` / `tournament` / `tournamentName`, **not** `competitionName` — Home cards can show "Competition unknown" even when the API sent `competitionName`. That is a **client mapping gap**, not a separate API test.

Edge states checked: empty Live; empty Results then fallback; canceled first row on team last; not-started at-bats empty; unknown ids 404; missing search q 400. **Not tested:** authenticated vs anonymous UI (keys are server-side); mobile viewport; `/endpoints` click-through as a user; Search typo; Manager navigation from a lineup click (lineup panel is raw JSON, not links).

K. Findings / defect register

Severity is product impact in this lab, not a vendor SLA.

ID	Severity	Area	Finding	Status
F-01	**Critical**	`/events` filters	`status=finished` and `inprogress` return total=0 while team `events/last` has thousands of games. Live and Results cannot be built from the global feed.	Open
F-02	**High**	List vs detail schema	List: string status + `{ series }` + `competitionName`. Detail: status object + `inningsBaseball` R/H/E + nested uniqueTournament. Home cannot print a baseball scoreboard from the list.	Open
F-03	**High**	Optional enrichments	`best-players` and team `rankings` 404 on a valid finished game / valid team. `top-performers` is 200. Same 404 class as unknown entity.	Open
F-04	**High**	Auth contract	`X-Live-Key` alone 200; `x-apim-key` alone 403; both 200; none 403 non-JSON. No written rule beyond setting both names.	Open
F-05	**High**	Catalog gap	Player season statistics path returns 200 with battingAvg/HR/AB and is not in the 42 routes or on the player page.	Open
F-06	**Medium**	`/h2h`	200 with null duels on a finished game.	Open
F-07	**Medium**	Standings home/away	200 empty `rows` vs a full total table.	Open
F-08	**Medium**	Pagination	`page > totalPages` is 200 empty with **stale high `total`**.	Open
F-09	**Medium**	`/fixtures/today` vs `/events`	Identical IDs and byte size in this run. Unclear whether "today" is a calendar slice.	Open
F-10	**Medium**	Tournament id on detail	`tournament.id` null; nested `uniqueTournament.id` present. Client prefers `tournament` so competition links break.	Open (API + client)
F-11	**Medium**	`/events?status=canceled`	total=0 while canceled events exist on team last.	Open
F-12	**Medium**	Client 404 copy	Any HTTP 404 is treated as "proxy not found". Enrichment 404s get the wrong operator hint.	Open (client)
F-13	**Low**	Error `message` type	Array vs string vs `{code,message}` (search 400 array confirmed).	Open
F-14	**Low**	Player/team preferred fields	Player `team` null; team `country` null on team resource.	Open

ID	Severity	Area	Finding	Status
F-15	**Low**	Featured-event semantics	Returns a finished game.	Open
F-16	**Low**	Lineup manager	Home `manager` null; away manager used for `/manager/{id}`.	Open
F-17	**Low**	`competitionName` unused	List rows include `competitionName`; client mapping does not read it → “Competition unknown”.	Open (client)

Lab timings (some list/detail calls over ~800ms) were recorded but are not promoted here as performance defects.

L. Usability / integration assessment

No numeric scores.

What integrates cleanly: search hits with `type` + `id` + `name`; event detail R/H/E; at-bats; lineups with batting order; player-statistics; umpires; weather; top-performers; team roster; media; near-events; competition total standings; seasons list; manager by id; pagination `hasMore` on directories.

What forces workarounds: Results tab hard-codes one team id; Home Live depends on a global inprogress filter that is empty; match cards must interpret string-or-object status and series-or-innings scores.

What looks like a broken page: best-players and rankings tabs (404 JSON); h2h tab (nulls); standings home/away (empty table JSON); Live (empty state).

Operator experience: config asks for both key names. The proxy sends both if present. Live-Key alone is enough on `/events?limit=1`; APIM alone is not. Customers copying only the APIM key will see 403.

Accessibility / visual QA: not in scope. Match extras are raw JSON, not baseball-specific widgets — fine for a lab, not a consumer UI.

M. Coverage summary (how counts were calculated)

Bucket	Count	How it was counted
Catalog routes	**42**	Official baseball catalog
Catalog routes executed	**42 / 42**	One GET per catalog template
Extra probes	**11**	finished/inprogress, paging, validation, standings splits, next events
Variation tests	**51**	States, second entities, deep pages, limits, status tokens
Full-catalog lab rows	**71**	Catalog plus error/paging rows; catalog covered 42
Recorded HTTP calls	**196**	181×200, 8×400, 7×404
UI screens with ≥1 recorded call	7 named screens + catalog sweep	**Manager** and **Endpoint lab page** have no screen rows
Additional GETs this review	Auth 4 header combos + path GETs in this report	GET only
Catalog cases in §G1 marked executed	42	All observed
§G1 PASS / FAIL / NEEDS INVESTIGATION	37 PASS, 3 FAIL (C-03, C-06, C-32), 2 NEEDS INVESTIGATION (C-12, C-17)	From §G1 Status column
Product journeys in §J	14 rows	One per journey; mix of PASS/FAIL/NOT TESTED
Load/stress	**0**	Explicitly not run
Destructive requests	**0**	GET/HEAD only

This is not 100% product certification. Gaps: Manager UI walk, `/endpoints` click-through, Search typo, Search player/competition on the Search **page** (only catalog audit), Team tabs beyond overview/featured/near/performance, Match tabs not all walked, incidents/injuries uncalled, POST 405 unexecuted, postponed filter without a known postponed id, whether inprogress=0 means no live baseball worldwide.

N. Recommendations and conclusion

Must

1. Make `/events?status=finished` and `inprogress` return the games that already exist on team/player `events/last` (F-01). Until then, Live/Results are not shippable without a team-specific hack.
2. Put detail-equivalent **status object** and **`inningsBaseball` (R/H/E)** on `/events` and `/fixtures/today` rows (F-02).
3. Return **200 + empty** for missing optional enrichments (`best-players`, team `rankings`) instead of 404, **or** drop them from the baseball catalog and point MOTM at `/top-performers` (F-03).

4. Publish one auth rule: which of `X-Live-Key` / `x-apim-key` is required (F-04).

Should

5. Add `GET /v1/baseball/player/{id}/unique-tournament/{tid}/season/{sid}/statistics` to the baseball catalog and to the player page (F-05) — it already returns battingAvg / HR / AB.
6. Populate `/h2h` or document null duels as the empty contract (F-06).
7. Fill standings `home`/`away` or remove those types from the product tabs (F-07).
8. For `page > totalPages`, either 400 or keep 200 but do not leave `total` looking like a full feed (F-08).
9. Document `/fixtures/today` vs `/events` (F-09).
10. Set `tournament.id` (or tell clients to use nested `uniqueTournament.id`); optionally prefer nested id when `tournament.id` is null (F-10).
11. Stop mapping API 404 bodies to “proxy not found” (F-12).

Nice-to-have

12. One `message` type on errors (F-13).
13. Player.team and team.country on profile resources (F-14).
14. Document featured-event vs near-events (F-15).
15. Manager id on both lineup sides; document roster size (F-16).
16. Map list `competitionName` in the client (F-17).
17. `hasMore`-only paging is already a reasonable client defense.

Conclusion

The Baseball Product is a thin, honest lab over **42** deployed GETs. Enough of the catalog works to demonstrate search, a finished match centre, a team roster, and a competition table. **Live and Results do not**, because the global `/events` status index does not match team history. List scoreboards are series-only; optional extras 404; season stats exist off-catalog. Treat this document as an evidence pack, not a go-live certificate.